

D609

Data Analytics at Scale

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A MapReduce job counts words across 2 TB of support tickets from a firm with offices in 9 cities. Two of the inputs are 'late order' and 'order late order.' Considering only these two inputs, what does the reducer output for the key 'order'?

- A. (order, 2)
- B. (order, 3)
- C. (order, 5)
- D. (order, [1, 1, 1])

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

In MapReduce, which phase moves all values for the same key to the same reducer?

- A. Combine
- B. Reduce
- C. Map
- D. Shuffle (and sort)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A data engineer at a 400-person insurer writes `df2 = df.filter(...)` and then `df3 = df2.select(...)` in Spark on a 30-node cluster. The Spark UI shows no jobs have run. Why?

- A. The filter failed silently.
- B. Spark runs each line immediately, so the UI must be broken.
- C. Transformations are lazy; nothing runs until an action such as `count()` or `write`.
- D. The cluster is too small to start a job.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Which of these Spark operations triggers execution?

- A. `filter()`
- B. `withColumn()`
- C. `count()`
- D. `groupBy()`, because any shuffle always runs immediately

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A Spark job at a music-streaming company with 90 million users runs map, then filter, then groupByKey on user ID. The job runs every hour. A new stage begins at the groupByKey. Why?

- A. Every transformation starts a new stage.
- B. filter is a wide transformation.
- C. The hourly schedule creates the stages.
- D. groupByKey is a wide transformation that shuffles data across partitions.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A Spark cluster at a logistics firm has 10 executors with 4 cores each (40 cores). A large dataset is read into only 8 partitions. The firm ships to all 50 states. What limits parallelism, and what fixes it?

- A. Fewer partitions are always faster.
- B. Too few executors; add more.
- C. Only 8 tasks can run at once; repartition to several times the core count.
- D. Use 50 partitions, one per state.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A stage has 200 partitions, the cluster has 40 cores, and each task takes 1 minute. What is the minimum stage time?

- A. 5 minutes
- B. 1 minute
- C. 200 minutes
- D. 8,000 minutes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An ML team in Toronto runs 50 iterations of gradient descent over the same 500 GB dataset in Spark, and each iteration re-reads the data from object storage. What change helps most?

- A. Switch to disk-based MapReduce.
- B. Cache or persist the dataset after the first read.
- C. Cache every intermediate DataFrame.
- D. Move the team closer to the data center.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

An analyst queries 3 of 200 columns from a 5 TB table stored as unpartitioned CSV, filtering to one event_date. The storage bucket sits in a region near Oregon. What is the best improvement?

- A. Move the bucket to another region.
- B. Convert to JSON.
- C. Gzip the CSV files.
- D. Store the table as Parquet partitioned by event_date.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A job takes 100 minutes on one node, and 20% of it is serial. The team moves it to a 50-node cluster that comes with a new cost dashboard. Assuming the rest parallelizes perfectly, what is the new runtime?

- A. 80.4 minutes
- B. 21.6 minutes
- C. 2 minutes
- D. 20 minutes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

If 25% of a job is serial, what is the maximum possible speedup with unlimited nodes?

- A. 4×
- B. Unlimited
- C. About 1.33×
- D. 75×

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A company’s on-premises Hadoop cluster, housed in a leased data center, must add compute nodes every time storage fills, even though CPU usage sits at 15%. Which cloud-native pattern addresses this?

- A. Keep data in object storage and scale compute clusters separately.
- B. Keep adding nodes as storage fills.
- C. Lease a different data center.
- D. Put bigger disks in each node.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A MapReduce job counts words across ~~2 TB of support tickets from a firm with offices in 9 cities~~. Two of the inputs are 'late order' and 'order late order.' Considering only these two inputs, what does the reducer output for the key 'order'?

Target word: reducer output **Decides:** 'order' appears once in the first and twice in the second.

A	Dropped element	Counts documents that contain the word, not occurrences.
B	Correct	Map emits (order, 1) three times; reduce sums to 3.
C	Wrong stem number	The total word count across both inputs.
D	Adjacent category	That is the grouped shuffle output before reduce.

2. Answer D Concept 1 · transfer

In MapReduce, which phase moves all values for the same key to the same reducer?

Target word: name the grouping phase **Decides:** Grouping by key across machines.

A	Confusion set	Pre-aggregates locally on each mapper.
B	Inverted operation	Aggregates after grouping has already happened.
C	Adjacent category	Emits pairs but does not group them.
D	Correct	Routes every key's values to one reducer.

3. Answer C Concept 2

A data engineer at a ~~400-person~~ insurer writes `df2 = df.filter(...)` and then `df3 = df2.select(...)` in Spark ~~on a 30-node cluster~~. The Spark UI shows no jobs have run. Why?

Target word: why nothing executed **Decides:** Only transformations have been called.

A	Confusion set	Nothing ran, so nothing failed.
B	Inverted operation	Spark defers execution.
C	Correct	Spark builds a plan and waits for an action.
D	Camouflage	Cluster size does not stop execution.

4. Answer C Concept 2 · transfer

Which of these Spark operations triggers execution?

Target word: identify an action **Decides:** Actions return results or write output.

A	Confusion set	A lazy transformation.
B	Adjacent category	A lazy transformation.
C	Correct	Returns a value, so the plan must run.
D	Absolute language	Wide transformations are lazy too.

5. Answer D Concept 3

A Spark job at a music-streaming company with 90 million users runs map, then filter, then groupByKey on user ID. The job runs every hour. A new stage begins at the groupByKey. Why?

Target word: why a stage boundary appears **Decides:** groupByKey needs data from all partitions.

A	Absolute language	Narrow transformations are pipelined into one stage.
B	Confusion set	filter is narrow; it works within a partition.
C	Camouflage	Scheduling does not define stages.
D	Correct	Shuffles split jobs into stages.

6. Answer C Concept 4

A Spark cluster at a logistics firm has 10 executors with 4 cores each (40 cores). A large dataset is read into only 8 partitions. The firm ships to all 50 states. What limits parallelism, and what fixes it?

Target word: parallelism bottleneck **Decides:** 8 partitions for 40 cores.

A	Absolute language	Fewer partitions reduce parallelism.
B	Dropped element	More cores cannot help with only 8 tasks.
C	Correct	One task per partition leaves 32 cores idle.
D	Camouflage	The state count does not decide partitioning.

7. Answer A Concept 4 · transfer

A stage has 200 partitions, the cluster has 40 cores, and each task takes 1 minute. What is the minimum stage time?

Target word: stage time from waves **Decides:** $200 / 40 = 5$ waves.

A	Correct	Five waves of 40 one-minute tasks.
B	Dropped element	Assumes all 200 tasks run at once.
C	Adjacent category	Runs tasks one at a time, ignoring the cores.
D	Inverted operation	Multiplies partitions by cores.

8. Answer B Concept 5

An ML team in Toronto runs 50 iterations of gradient descent over the same 500 GB dataset in Spark, and each iteration re-reads the data from object storage. What change helps most?

Target word: speed up iterative work **Decides:** The same data is reused 50 times.

A	Inverted operation	MapReduce writes to disk between steps and is slower for iteration.
B	Correct	Later iterations read from memory.
C	Absolute language	Over-caching wastes memory; cache what is reused.
D	Camouflage	Team location does not affect cluster I/O.

9. Answer D Concept 6

An analyst queries 3 of 200 columns from a 5 TB table stored as unpartitioned CSV, filtering to one event_date. ~~The storage bucket sits in a region near Oregon.~~ What is the best improvement?

Target word: storage layout fix **Decides:** Few columns and a date filter.

A	Camouflage	Region does not reduce the data scanned.
B	Confusion set	Still row-based and larger.
C	Adjacent category	Smaller files, but every row and column is still read.
D	Correct	Reads only 3 columns and only the matching date partition.

10. Answer B Concept 7

A job takes 100 minutes on one node, and 20% of it is serial. The team moves it to a 50-node cluster ~~that comes with a new cost dashboard.~~ Assuming the rest parallelizes perfectly, what is the new runtime?

Target word: runtime with Amdahl's law **Decides:** Serial 20 minutes; parallel 80 minutes / 50.

A	Inverted operation	Parallelizes the serial part instead: $80 + 20 / 50$.
B	Correct	$20 + 80 / 50 = 20 + 1.6$.
C	Dropped element	Divides the whole job by 50, ignoring the serial part.
D	Adjacent category	The limit with unlimited nodes.

11. Answer A Concept 7 · transfer

If 25% of a job is serial, what is the maximum possible speedup with unlimited nodes?

Target word: speedup ceiling **Decides:** $1 / \text{serial fraction}$.

A	Correct	$1 / 0.25 = 4$.
B	Absolute language	The serial part caps speedup.
C	Inverted operation	$1 / 0.75$ uses the parallel fraction.
D	Wrong stem number	Reads the parallel share as a speedup.

12. Answer A Concept 8

A company's on-premises Hadoop cluster, ~~housed in a leased data center,~~ must add compute nodes every time storage fills, even though CPU usage sits at 15%. Which cloud-native pattern addresses this?

Target word: architecture pattern **Decides:** Storage and compute are tied together.

A	Correct	Each resource grows only when it is needed.
B	Dropped element	Continues paying for idle CPUs.
C	Camouflage	The location is not the constraint.
D	Adjacent category	Still ties storage growth to cluster nodes.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	MapReduce phases	Map emits key-value pairs; shuffle groups all values for a key onto one reducer; reduce aggregates each key's values.	Counting documents instead of occurrences, or thinking map or reduce does the grouping.	A word-count style job or a question about which phase moves data.
2	Lazy evaluation in Spark	Transformations (filter, select, groupBy) only build a plan; nothing executes until an action (count, collect, write).	Expecting each line of Spark code to run immediately.	Code has run but no jobs appear.
3	Narrow vs. wide transformations	Narrow transformations (map, filter) work within a partition; wide ones (groupBy, join) shuffle data across the cluster and start a new stage.	Thinking every transformation shuffles, or that filter is wide.	A stage boundary at a grouping or join.
4	Partitions and parallelism	Each partition is one task; parallelism is capped by the number of partitions and cores, and stage time $\approx$ number of task waves $\times$ task time.	Adding executors when there are fewer partitions than cores.	Partition count far below core count, or tasks, cores, and task time.
5	Caching for iterative work	Persist a dataset in memory when it is reused across many iterations; this is Spark's main edge over disk-based MapReduce.	Re-reading from storage every iteration, or caching everything indiscriminately.	Many passes over the same data.
6	Columnar formats and pruning	Columnar formats like Parquet read only the needed columns, and partitioning by a filter column lets the engine skip whole partitions.	Compressing row-based files and expecting column-level savings.	A few columns and a date filter over a huge row-format table.
7	Amdahl's law	Runtime = serial part + parallel part / nodes; maximum speedup with unlimited nodes = $1 / \text{serial fraction}$.	Dividing the whole job by the node count.	A serial percentage and a node count.
8	Separating storage and compute	Cloud-native architectures keep data in object storage and scale compute clusters independently, so each grows only when needed.	Adding compute nodes just to gain storage.	Storage full while CPUs sit idle.

Course scope (public description): Data Analytics at Scale builds on previous data engineering courses and discusses approaches for analyzing large data sets. The course discusses map/reduce approaches, Apache Spark, and cloud-native solutions for developing, automating, and scaling data analytics. Also discussed are methods for integrating data processing pipelines and data stores to create comprehensive data analytics architectures.